对偶

	原问题	对偶问题
目标. (LP)	max	min

原问题 变量数 $\Longleftrightarrow$ 对偶问题 约束数

原问题 约束数 $\Longleftrightarrow$ 对偶 变量数

Thm (弱对偶定理) $c^Tx \le b^Ty$ x, y 可行

Thm (强对偶定理) 若原问题存在最优解, 则对偶问题也存在最优解, 且两者目标值相等.

Thm (互补松弛定理, complementarity slackness theorem)

对于LP问题, 若 x, y 对应原问题, 对偶问题的可行解, x_s, y_s 对应原, 对偶问题的松弛变量, 则 x, y 为最优解, 当且仅当:

① $y^Tx_s = 0$

② $x^Ty_s = 0$

Proof: Primal: $\max\ c^Tx$ s.t. $Ax + Ix_s = b$, $x, x_s \ge 0$

Dual: $\min\ b^Ty$ s.t. $A^Ty - Iy_s = c$, $y, y_s \ge 0$

$\Rightarrow c^Tx = (A^Ty - Iy_s)^Tx = y^TAx - y_s^Tx$

$b^Ty = (Ax + Ix_s)^Ty = x^TA^Ty + x_s^Ty$

由于 ①, ②, 可推出 $c^Tx = b^Ty$, 根据强对偶定理, x, y 为最优解.

注: 互补性 (complementarity) 指: $\{a \ge 0, b \ge 0, a^Tb = 0\}$

E.g.:

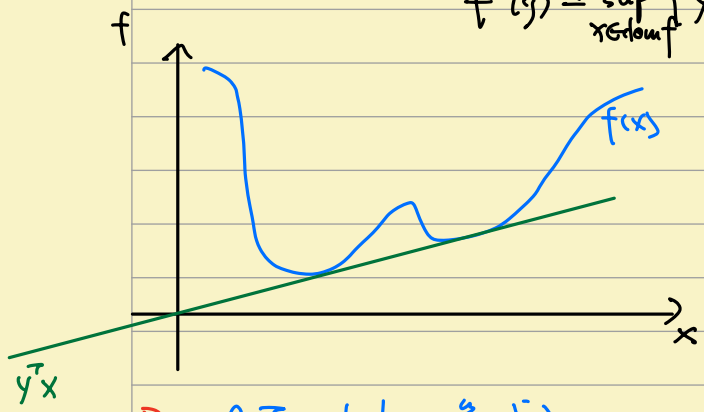
① $A^Ty - c \ge 0$, $x \ge 0$, $(A^Ty - c)^Tx = 0$

② $b - Ax \ge 0$, $y \ge 0$, $(b - Ax)^Ty = 0$

共轭函数

Defn: (共轭函数) 若 $f(x)$ 是一个适当函数，其共轭函数定义为

$$f^*(y) = \sup_{x \in \mathrm{dom} f} \{ y^T x - f(x) \}$$



Prop (Fenchel 不等式)

$$f(x) + f^*(y) \geq x^T y$$

Proof:

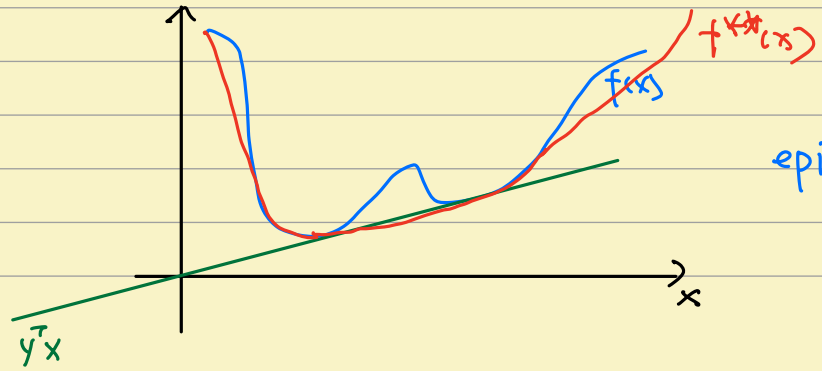
$$f^*(y) = \sup_{x \in \mathrm{dom} f} \{ y^T x - f(x) \} \geq y^T x - f(x)$$

Defn (二次共轭函数) 若 $f(x)$ 是一个适当函数，其二次共轭函数为

$$f^{**}(x) = \sup_{y \in \mathrm{dom} f^*} \{ x^T y - f^*(y) \}$$

Note: 根据 Fenchel 不等式，有

$$\left.\begin{array}{l} f(x) + f^*(y) \geq x^T y \\ f^*(y) + f^{**}(x) \geq y^T x \end{array}\right\} \Rightarrow f^{**}(x) \leq f(x)$$



$\mathrm{epi} f \subseteq \mathrm{epi} f^{**}$

Thm: 若 f 是一个闭凸函数，则

$$f^{**}(x) = f(x) \quad . \forall x \in \mathrm{dom} f.$$

E.g.: $f(x) = \frac{1}{2}x^TAx + b^Tx + c$ 求 f 的共轭 $(A \succeq 0)$

$$f^*(y) = \sup_{x \in \mathrm{dom} f} \{y^Tx - f(x)\}$$

$$= \sup_{x \in \mathrm{dom} f} \{y^Tx - \frac{1}{2}x^TAx - b^Tx - c\}$$

令 $g(x) = y^Tx - \frac{1}{2}x^TAx - b^Tx - c$

$$\nabla g(x) = y - Ax - b = 0 \Rightarrow x^* = A^{-1}(y-b)$$

$$\therefore f^*(y) = y^Tx^* - f(x^*)$$

$$= y^TA^{-1}(y-b) - \frac{1}{2}(y-b)^TA^{-1}AA^{-1}(y-b) - b^TA^{-1}(y-b) - c$$

$$= \frac{1}{2}(y-b)^TA^{-1}(y-b) - c$$

E.g.: $f(x) = \|x\|_2$ 求 f 的共轭.

$$f^*(y) = \sup_{x \in \mathrm{dom} f} \{y^Tx - f(x)\}$$

令 $g(x) = y^Tx - \|x\|_2$，定义对偶范数 $\|y\|_* = \sup_{\|x\| \le 1} x^Ty$

① 若 $\|y\|_2 \le 1$，$y^Tx \le \|y\|_2 \|x\|_2 \le \|x\|_2$，且 $x=0$ 时等号成立

$$\Rightarrow \sup_{x \in \mathrm{dom} f} \{y^Tx - \|x\|_2\} = 0$$

② 若 $\|y\|_2 > 1$，$\sup_{x \in \mathrm{dom} f} \{y^Tx - \|x\|_2\}$ $\quad P_y x = t \cdot \frac{y}{\|y\|_2}$

$$= \sup_{t \in \mathbb{R}} t(\|y\|_2 - 1) = +\infty$$

指示函数

$$\longrightarrow \quad f^*(y) = \begin{cases} 0 & \|y\|_2 \le 1 \\ +\infty & \|y\| > 1 \end{cases}$$

Note：① $\|\cdot\|_2$ 的对偶范数 $\|\cdot\|_2$

② 范数的共轭函数为其对偶范数的单位球的指示函数

一般约束优化问题的对偶问题

一般约束优化问题

$$(P) \quad \begin{cases} \min\limits_{x \in \mathbb{R}^n} f(x) \\ \text{s.t.} \quad c_1(x) \le 0 \Rightarrow c_{1i}(x) \le 0,\ i \in I \quad |I| = m \\ \qquad\ \ c_2(x) = 0 \Rightarrow c_{2j}(x) = 0,\ j \in \mathcal{E} \quad |\mathcal{E}| = p \end{cases}$$

指标集　　m个不等式约束

最优值 $z^* = \min\limits_{x \in \Omega} f(x)$

可行域：$\Omega = \{x \in \mathbb{R}^n \mid c_1(x) \le 0,\ c_2(x) = 0\}$

拉格朗日函数 (Lagrangian function)

$\lambda \in \mathbb{R}^m$

$\mu \in \mathbb{R}^p$

$$L(x;\lambda,\mu) = f(x) + \lambda^T c_1(x) + \mu^T c_2(x)$$
$$= f(x) + \sum_{i \in I} \lambda_i c_{1i}(x) + \sum_{j \in \mathcal{E}} \mu_j c_{2j}(x)$$

Defn (Lagrangian 对偶函数) Lagrangian 对偶函数 $g: \mathbb{R}^m_+ \times \mathbb{R}^p \to [-\infty, +\infty)$ 是 Lagrangian 函数 $L(x;\lambda,\mu)$ 对于 $\lambda \ge 0$, $\mu \in \mathbb{R}^p$ 取下界：

$$g(\lambda,\mu) = \inf_{x \in \mathbb{R}^n} L(x;\lambda,\mu)$$

E.g.:

$$\begin{cases} \max\ c^T x \\ \text{s.t.}\ Ax = b \\ \qquad x \ge 0 \end{cases} \Rightarrow \begin{cases} \min\ b^T y \\ \text{s.t.}\ A^T y \ge c \\ \qquad y \ge 0 \end{cases}$$

$$\min \ -c^T x$$
$$\text{s.t.} \ -x \le 0$$
$$Ax = b$$

Lagrangian 函数：

$$L(x;\lambda,\mu) = -c^T x + \lambda^T(-x) + \mu^T(Ax-b)$$
$$= (A^T\mu - \lambda - c)^T x - b^T\mu$$

Lagrangian 对偶函数

$$g(\lambda,\mu) = \inf_{x\in\mathbb{R}^n} L(x;\lambda,\mu)$$

① $A^T\mu - \lambda - c = 0 \Rightarrow \inf_{x\in\mathbb{R}^n} L(x;\lambda,\mu) = L(x;\lambda,\mu) = -b^T\mu$

② $A^T\mu - \lambda - c \neq 0$

if $A^T\mu - \lambda - c > 0$, take $x = -\infty$

$\Rightarrow \inf_{x\in\mathbb{R}^n} L(x;\lambda,\mu) = -\infty$

if $A^T\mu - \lambda - c < 0$, take $x = +\infty$

$\Rightarrow \inf_{x\in\mathbb{R}^n} L(x;\lambda,\mu) = -\infty$

$$\therefore \ g(\lambda,\mu) = \inf_{x\in\mathbb{R}^n} L(x;\lambda,\mu) = \begin{cases} -b^T\mu & \text{if } A^T\mu - \lambda - c = 0 \\ -\infty & \text{otherwise.} \end{cases}$$

Lagrangian 对偶问题等价写成

$$\max \ -b^T\mu$$
$$\text{s.t} \ A^T\mu - c - \lambda = 0$$
$$\lambda \ge 0$$

$$\Rightarrow \min \ b^T\mu$$
$$\text{s.t} \ A^T\mu - c \ge 0$$

$$\min \ b^T y$$
$$\text{s.t.} \ A^T y \ge c$$
$$y \ge 0$$

$$\min_{x\in\mathbb{R}^n} f(x)$$
$$\text{s.t.}\quad c_1(x)\le 0,\quad c_2(x)=0$$

Thm (Weak duality theorem) 对于任意 $\lambda\ge 0$, $\mu\in\mathbb{R}^p$, 有

$$g(\lambda,\mu)\le z^* \longrightarrow \text{原问题的最优值}$$

Proof: 若 $\bar{x}\in\Omega$, 则 $c_1(\bar{x})\le 0$, $c_2(\bar{x})=0$

$$\mathcal{L}(\bar{x};\lambda,\mu)=f(\bar{x})+\cancel{\lambda^T c_1(\bar{x})}+\cancel{\mu^T c_2(\bar{x})}\le f(\bar{x})$$

$\lambda^T \ge 0$, $c_1(\bar{x})\le 0$, $\mu^T\ge 0$, $c_2(\bar{x})=0$

有 $g(\lambda,\mu):=\inf_{x\in\mathbb{R}^n}\mathcal{L}(x;\lambda,\mu)\le\mathcal{L}(\bar{x};\lambda,\mu)\le f(\bar{x})\quad\forall x\in\Omega$

对 $\bar{x}$ 取下界 $f(\bar{x})=\min_{x\in\Omega} f(x)=z^*$

$$\Rightarrow g(\lambda,\mu)\le z^*$$

Lagrangian 对偶问题

$$\max\ g(\lambda,\mu)$$
$$\text{s.t.}\ \lambda\ge 0,\ \mu\in\mathbb{R}^p$$

对偶变量: λ,μ

最优值: w^* → w^* 是 z^* 最优下界.

weak duality inequality

Prop: ① 对于 Lagrangian 对偶问题, $z^*\ge w^*$ 一定成立

② 对偶问题一定为凸问题 ← 凸问题一般易于求解

① 目标函数为凸

② 可行域为凸

B.g.: 求下列问题的 Lagrangian 对偶问题

$$\begin{aligned}\max\ & c^Tx\\ \text{s.t.}\ & A_1x=b_1\\ & A_2x\le b_2\\ & A_3x\ge b_3\\ & x\ge 0\end{aligned}\quad\Rightarrow\quad\begin{aligned}\min\ & -c^Tx\\ \text{s.t.}\ & \begin{pmatrix}A_2\\ -A_3\\ -I\end{pmatrix}x\le\begin{pmatrix}b_2\\ -b_3\\ 0\end{pmatrix}\begin{matrix}\lambda_1\\ \lambda_2\\ \lambda_3\end{matrix}\\ & A_1x=b_1\quad\mu\end{aligned}$$

令 $A=\begin{pmatrix}A_2\\ -A_3\\ -I\end{pmatrix}$ $b=\begin{pmatrix}b_2\\ -b_3\\ 0\end{pmatrix}$ $\Rightarrow$

$$\begin{aligned}\min\ & -c^Tx\\ \text{s.t}\ & Ax\le b\\ & A_1x=b_1\end{aligned}$$

$$\mathcal{L}(x;\lambda_1,\lambda_2,\lambda_3,\mu) = -c^T x + \lambda_1^T(A_2 x - b_2) + \lambda_2^T(-A_3 x + b_3) + \lambda_3^T(-x) + \mu^T(A_1 x - b_1)$$

$$= \left(A_2^T\lambda_1 - A_3^T\lambda_2 + A_1^T\mu - \lambda_3 - c\right)^T x - b_2^T\lambda_1 + b_3^T\lambda_2 - b_1^T\mu$$

$$g(\lambda_1,\lambda_2,\lambda_3,\mu) = \inf_{x\in\mathbb{R}^n} = \left(A_2^T\lambda_1 - A_3^T\lambda_2 + A_1^T\mu - \lambda_3 - c\right)^T x - b_2^T\lambda_1 + b_3^T\lambda_2 - b_1^T\mu$$

$$= \begin{cases} -b_2^T\lambda_1 + b_3^T\lambda_2 - b_1^T\mu & \text{if } A_2^T\lambda_1 - A_3^T\lambda_2 + A_1^T\mu - \lambda_3 - c = 0 \\ -\infty & \text{otherwise} \end{cases}$$

等价写成:

$$\begin{aligned} \max \quad & -b_2^T\lambda_1 + b_3^T\lambda_2 - b_1^T\mu \\ \text{s.t.} \quad & A_2^T\lambda_1 - A_3^T\lambda_2 + A_1^T\mu - \lambda_3 - c = 0 \\ & \lambda_{1-3} \geq 0 \end{aligned}$$

E.g.: 写出下面问题的 Lagrangian 对偶问题

$$\begin{aligned} \min \quad & \|x\|_2 \\ \text{s.t.} \quad & Ax = b \end{aligned}$$

$$\mathcal{L}(x;\mu) = \|x\|_2 - \mu^T(Ax - b)$$

$$g(\mu) = \inf_{x\in\mathbb{R}^n}\left\{\|x\|_2 - \mu^T(Ax-b)\right\}$$

$$= \inf_{x\in\mathbb{R}^n}\left\{\|x\|_2 - (A^T\mu)^T x\right\} + b^T\mu$$

$$= \sup_{x\in\mathbb{R}^n}\left\{(A^T\mu)^T x - \|x\|_2\right\} + b^T\mu = \begin{cases} b^T\mu & \text{if } \|A^T\mu\|_2 \leq 1 \\ -\infty & \text{otherwise} \end{cases}$$

$$\Rightarrow \begin{cases} \max \quad b^T\mu \\ \text{s.t.} \quad \|A^T\mu\|_2 \leq 1 \end{cases}$$

E.g.:

$$\begin{aligned} \min_x \quad & x^T W x \qquad W \succ 0 \\ & Ax \leq b \end{aligned}$$

$$\mathcal{L}(x;\lambda) = x^T W x + \lambda^T(Ax - b)$$

$$g(\lambda) = \inf_{x\in\mathbb{R}^n}\left\{x^T W x + \lambda^T(Ax - b)\right\}$$

$$= \inf_{x \in \mathbb{R}^n} \left\{ x^T W x + (A^T\lambda)^T x \right\} - b^T\lambda$$

令 $h(x) = x^T W x + (A^T\lambda)^T x$

$$\nabla h(x) = 2Wx + A^T\lambda = 0 \Rightarrow x^* = -\frac{1}{2} W^{-1} A^T \lambda$$

$$g(\lambda) = (x^*)^T W x^* + (A^T\lambda)^T x^* - b^T\lambda$$

$$= -\frac{1}{4} \lambda^T A W^{-1} A^T \lambda - b^T \lambda$$

$$\Rightarrow \begin{cases} \max & -\frac{1}{4} \lambda^T A W^{-1} A^T \lambda - b^T \lambda \\ \text{s.t} & \lambda \geq 0 \end{cases}$$

对偶问题的解释:

见补充 Note

3. 对偶最优解 $\quad y = (y_1, y_2) = (0, -2)$

$$y_1 \Rightarrow -x_1 + x_2 + x_3 + x_{s_1} = 4$$
$$y_2 \Rightarrow -x_1 + x_2 - x_3 + x_{s_2} \leq 6$$

CST $\Rightarrow y^T x_s = 0 \qquad y_2 = -2 \neq 0 \Rightarrow x_{s_2} = 0$

$$x_{s_2} = 0 \Rightarrow -x_1 + x_2 - x_3 = 6$$

$$\therefore \begin{cases} -x_1 + x_2 + x_3 = 4 \\ -x_1 + x_2 - x_3 = 6 \end{cases} \Rightarrow x_3 = -1$$

$$-x_1 + x_2 = 5$$

$$x_1 \leq 0$$
$$x_2 \geq 0$$

$$x_2 = 5 + x_1$$

$$\min \quad z = 2x_1 - x_2 + 2x_3$$
$$= 2x_1 - x_2 - 2$$
$$= 2x_1 - x_1 - 5 - 2$$
$$= x_1 - 7$$
$$x_1 = -5 \qquad x_2 = 0$$

4. $y = (0.8, 0.6)$

$\min z = 2x_1 + 3x_2 + 5x_3 + 2x_4 + 3x_5$

① $\Rightarrow \begin{cases} x_1 + x_2 + 2x_3 + x_4 + 3x_5 = 4 \\ 2x_1 - x_2 + 3x_3 + x_4 + x_5 = 3 \end{cases}$

$$\begin{array}{ll} \max & y^T b \\ \text{s.t} & A^T y \le c \\ & y \ge 0 \end{array} \quad \Leftarrow \quad \begin{array}{ll} \min & c^T x \\ \text{s.t} & Ax \ge b \\ & x \ge 0 \end{array}$$

② 互补性： $c - A^T y \ge 0 \quad x \ge 0, \ (c - A^T y)^T x = 0$

$$A^T y^* = \begin{bmatrix} 1 & 2 \\ 1 & -1 \\ 2 & 3 \\ 1 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix} = \begin{bmatrix} 2 \\ 0.2 \\ 3.4 \\ 1.4 \\ 3 \end{bmatrix} \begin{matrix} = \\ < \\ < \\ < \\ = \end{matrix} \quad c = \begin{bmatrix} 2 \\ 3 \\ 5 \\ 2 \\ 3 \end{bmatrix}$$

$\Rightarrow x_2 = x_3 = x_4 = 0$

$\Rightarrow \begin{cases} x_1 + 3x_5 = 4 \\ 2x_1 + x_5 = 3 \end{cases} \Rightarrow \begin{cases} x_1^* = 1 \\ x_2^* = 1 \end{cases}$

$x^* = (1, 0, 0, 0, 1)$